

5.5

$$\begin{aligned}
 a. \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T} \\
 &= \frac{T(T+1)(2T+1)}{6} - \frac{(\frac{T(T+1)}{2})^2}{T} \\
 &= \frac{T(T+1)(2T+1)}{6} - \frac{T^2(T+1)^2}{4T} \\
 &= \frac{2T(T+1)(2T+1) - 3T(T+1)^2}{12} \\
 &= \frac{T(T+1)(4T+2-3T-3)}{12} \\
 &= \frac{T(T+1)(T-1)}{12} \\
 &= \frac{T(T^2-1)}{12}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{var}(b_2|x) &= \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2} \\
 &= \frac{\sigma^2}{\frac{T(T^2-1)}{12}} \\
 &= \frac{12\sigma^2}{T(T^2-1)}
 \end{aligned}$$

As $T \rightarrow \infty$, $\text{var}(b_2|x) = 0$, the least squares estimator is consistent.

$$b. x_t = 0.5^t$$

$$\sum_{t=1}^T x_t = \frac{0.5(1-0.5^T)}{1-0.5} = 1 - 0.5^T$$

$$\sum_{t=1}^T x_t^2 = \sum_{t=1}^T (0.25)^t = \frac{0.25(1-0.25^T)}{1-0.25} = \frac{1}{3}(1-0.25^T)$$

$$\begin{aligned}
 \sum_{t=1}^T (x_t - \bar{x})^2 &= \sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T} \\
 &= \frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}
 \end{aligned}$$

$$\lim_{T \rightarrow \infty} \sum_{t=1}^T (x_t - \bar{x})^2 = \frac{1}{3}(1-0) - 0 = \frac{1}{3}$$

$$\lim_{T \rightarrow \infty} \text{var}(b_2|x) = \frac{\sigma^2}{\frac{1}{3}} = 3\sigma^2$$

Since $\lim_{T \rightarrow \infty} \text{var}(b_2|x) \neq 0$, the least squares estimator b_2 is not consistent.

c. 在(a)中, 随着 T 增加, 會提供越來越多關於 β_2 以及 x 和 y 之間的資訊, 估計會越來越精準。

在(b)中, $x_t = (0.5)^t$ 迅速趨近於 0。即使再增加 T , 也不會提供更多新的 β_2 資訊。

$$5.6 \quad df = 63 - 3 = 60, \quad t_{0.025; 60} \approx 2$$

$$a. H_0: \beta_2 = 0 \quad \text{vs} \quad H_1: \beta_2 \neq 0$$

$$b_2 = 3$$

$$\hat{\text{var}}(b_2) = 4 \Rightarrow \text{se}(b_2) = 2$$

$$t_c = \frac{b_2 - 0}{\text{se}(b_2)} \sim t_{(n-3)}$$

$$t^* = \frac{3-0}{2} = 1.5$$

Rejection region: $|t^*| > t_c = 2$

$\therefore |t^*| = 1.5 < t_c$, we do not reject H_0 .

b. $H_0: \beta_1 + 2\beta_2 = 5$ vs $H_1: \beta_1 + 2\beta_2 \neq 5$

$$b_1 + 2b_2 = 2 + 2(3) = 8, \quad \widehat{\text{var}}(b_1 + 2b_2) = \widehat{\text{var}}(b_1) + 2^2 \widehat{\text{var}}(b_2) + 2(2) \widehat{\text{cov}}(b_1, b_2) = 3 + 4(4) + 4(-2) = 11$$

$$t_{\text{stat}} = \frac{(b_1 + 2b_2) - 5}{\text{se}(b_1 + 2b_2)} \sim t_{(N-3)}$$

$$t^* = \frac{8-5}{\sqrt{11}} = 0.9045$$

$\therefore$ Since $|t^*| = |0.9045| < 2$, we do not reject H_0 .

c. $H_0: \beta_1 - \beta_2 + \beta_3 = 4$ vs $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

$$b_1 - b_2 + b_3 = 2 - 3 - 1 = -2$$

$$\begin{aligned} \widehat{\text{var}}(b_1 - b_2 + b_3) &= \widehat{\text{var}}(b_1) + \widehat{\text{var}}(b_2) + \widehat{\text{var}}(b_3) - 2\widehat{\text{cov}}(b_1, b_2) + 2\widehat{\text{cov}}(b_1, b_3) - 2\widehat{\text{cov}}(b_2, b_3) \\ &= 3 + 4 + 3 - 2(-2) + 2(1) - 2(0) \\ &= 16 \end{aligned}$$

$$t_{\text{stat}} = \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} \sim t_{(N-3)}$$

$$t^* = \frac{-2-4}{\sqrt{16}} = \frac{-6}{4} = -1.5$$

$\therefore$ Since $|t^*| = |-1.5| < 2$, we do not reject H_0 .

5.20 a.
$$\begin{aligned} \hat{\beta}_{2, \text{mean}} &= \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} \\ &= \frac{\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2 - \beta_1 - \beta_2 \bar{x}_1 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \\ &= \frac{\beta_2(\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \\ &= \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \end{aligned}$$

ML4: $E(e_i | X) = 0$

$$E[\hat{\beta}_{2, \text{mean}} | X] = \beta_2 + \frac{E(\bar{e}_2) - E(\bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + 0 = \beta_2$$

$\therefore \hat{\beta}_{2, \text{mean}}$ is unbiased.

$$\begin{aligned} b. \text{var}(\hat{\beta}_{2, \text{mean}} | X) &= \text{var}\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \mid X\right) \\ &= 0 + \text{var}\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \mid X\right) \\ &= \frac{\text{var}(\bar{e}_2 - \bar{e}_1 | X)}{(\bar{x}_2 - \bar{x}_1)^2} \end{aligned}$$

Each half contains $N/2$ observations:

$$\text{var}(\bar{e}_2 | X) = \frac{\sigma^2}{N/2} = \frac{2\sigma^2}{N}, \quad \text{var}(\bar{e}_1 | X) = \frac{2\sigma^2}{N}$$

$$\text{var}(\hat{\beta}_{2, \text{mean}} | X) = \frac{2\sigma^2/N + 2\sigma^2/N}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

$$\begin{aligned} c. \text{var}(\hat{\beta}_{2, \text{mean}}) &= E[\text{var}(\hat{\beta}_{2, \text{mean}} | X)] - \text{var}(E[\hat{\beta}_{2, \text{mean}} | X]) \\ &= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] - \text{var}(\beta_2) = 0 \\ &= \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \end{aligned}$$

d. Consistent requires:

$$\text{var}(\hat{\beta}_{2,\text{mean}} | x) = 0$$

$$\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} = 0$$

$$\Rightarrow N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$$

When $\bar{x}_2 - \bar{x}_1$ converges to a positive constant, the variance is of order $1/N$, and approach 0 so the estimator is consistent.

e. (i) $\text{var}(b_2 | x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$

$$\textcircled{2} \text{var}(\hat{\beta}_{2,\text{mean}} | x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

(ii) $\textcircled{1} E[(S_x^2)^{-1}]$, $E[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}]$

$$S_x^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$

> print(results)

	N	Estimate_var_b2	Estimate_var_b2mean	Estimate_inv_sx2	Estimate_term_mean
1	100	1.22102695554	1.64733901227	0.1221026956	0.1647339012
2	500	0.24102733893	0.32215915932	0.1205136695	0.1610795797
3	1000	0.12017834737	0.16039220582	0.1201783474	0.1603922058
4	5000	0.02400742914	0.03202002249	0.1200371457	0.1601001124

f. $E[\text{var}(\hat{\beta}_{2,\text{mean}} | x)] > E[\text{var}(b_2 | x)]$

$\hat{\beta}_{2,\text{mean}}$ has larger variance than OLS estimator b_2 .

Both variances decrease as N increases, roughly at the rate $1/N$, both estimators are consistent, but OLS is more efficient.

g. $x \sim U(0, 10)$, $\text{Var}(x) = \frac{(10-0)^2}{12} = \frac{100}{12} = \frac{25}{3}$

$$S_x^2 \xrightarrow{P} \frac{25}{3}$$

$$\therefore (S_x^2)^{-1} \xrightarrow{P} 0.12$$

lower half: $(0, 5)$

upper half: $(5, 10)$

As N becomes large, $\bar{x}_1 \xrightarrow{P} E[U(0, 5)] = \frac{0+5}{2} = 2.5$

$$\bar{x}_2 \xrightarrow{P} E[U(5, 10)] = \frac{5+10}{2} = 7.5$$

$$\Rightarrow \bar{x}_2 - \bar{x}_1 \xrightarrow{P} 5$$

$$\therefore \frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{P} \frac{4}{25} = 0.16$$

h. As N becomes large, both sample means converge to same population 5.

$$\bar{x}_1 \xrightarrow{P} 5, \bar{x}_2 \xrightarrow{P} 5$$

$$\Rightarrow \bar{x}_2 - \bar{x}_1 \xrightarrow{P} 0$$

$$\text{var}(\hat{\beta}_{2,\text{mean}} | x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

Consistency requires $N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$

Under randomly splitting, $\bar{x}_2 - \bar{x}_1$ converges to 0, therefore $\hat{\beta}_{2,\text{mean}}$ is not consistent.